

ECEN-629: Homework 1

Due on September 16, 2025 at 12:45 pm

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Problem 1

Compute the norms of the following vector $x = (-100, -99, -98, \dots, 99, 100)^T$.

- (a) ℓ_1 norm;
- (b) ℓ_∞ norm;
- (c) ℓ_2 norm.

Solution

We have the definition of a norm as

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

Then for each part we have:

$$\begin{aligned} \ell_1 : \|x\|_1 &= |-100| + |-99| + \dots + |99| + |100| \\ &= 2(1 + 2 + \dots + 99 + 100) \\ &= 2 \cdot \frac{100 \cdot 101}{2} \\ &= 10100 \end{aligned}$$

$$\begin{aligned} \ell_\infty : \|x\|_\infty &= \max\{|x|\} \\ &= 100 \end{aligned}$$

$$\begin{aligned} \ell_2 : \|x\|_2 &= \sqrt{(-100)^2 + (-99)^2 + \dots + (99)^2 + (100)^2} \\ &= \sqrt{2(1^2 + 2^2 + \dots + 99^2 + 100^2)} \\ &= \sqrt{2 \cdot \frac{100 \cdot 101 \cdot 201}{6}} \\ &= \sqrt{676700} \\ &\approx 822.62 \end{aligned}$$

Problem 2

For a quadratic function $f(x) = x^T P x + q^T x + r$ where P is not necessarily symmetric, show that the gradient is $\nabla f(x) = (P + P^T)x + q$. Find the Hessian of the function. Provide details in your proofs.

Solution

Problem 3

A matrix $A \in \mathbb{R}^{m \times n}$ can also be viewed as a vector, by stacking all its n columns into a single length- mn column. Let $x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$ be two column vectors. Show that

$$\text{tr}(A^T xy^T) = \langle xy^T, A \rangle = x^T Ay,$$

where tr is the trace of the matrix.

Solution

Problem 4

Let $B \in \mathbb{R}^{n \times n}$ be a given symmetric matrix of size n -by- n . Determine if the following set is convex: the set of all n -by- n symmetric A matrices, such that $A - B$ is positive semidefinite. If it is convex, provide your reasoning; if not, provide a counter-example.

Solution

Problem 5

A set is defined as

$$\{x \in \mathbb{R}^n \mid \|x - a\|_2 \leq \theta \|x - y\|_2 \text{ for all } y \in S\}$$

where $S \subseteq \mathbb{R}^n$.

- (a) Determine whether the set is convex, which may depend on the value of θ . For the case that it is, provide your reasoning; otherwise, provide a counter-example.
- (b) Let $a = (1, 0)^T$ and $S = \{(-1, 0)^T, (0, -1)^T\}$. For $\theta = 2$ and $\theta = 0.5$, sketch the resultant set, respectively.

Solution

Problem 6

- (a) Show that if S_1 and S_2 are convex set in $\mathbb{R}^n$, then the set sum

$$S \triangleq \{x_1 + x_2 \mid x_1 \in S_1, x_2 \in S_2\}$$

is also convex.

- (b) Let $S_1 = \{(-1, -1)^T\}$ and $S_2 = \{(2, 2)^T + u \mid \|u\| \leq 1\}$. What is their set sum?
- (c) Show that if S_1 and S_2 are convex set in $\mathbb{R}^n$, then the inverse sum

$$S \triangleq \bigcup_{1 \geq \lambda \geq 0} \{\lambda S_1 \cap (1 - \lambda) S_2\}$$

is also convex, where $\lambda S_j = \{\lambda x_j \mid x_j \in S_j\}$, $j = 1, 2$.

- (d) Let $S_1 = \{(1, 1)^T\}$ and $S_2 = \{(2, 2)^T + u \mid \|u\| \leq 1\}$. What is their inverse sum?

Solution

Problem 7

It is known that one halfspace contains another in the following form

$$\{x \mid a^T x \leq b\} \subseteq \{x \mid c \cdot a^T x \leq \tilde{b}\}.$$

Provide the conditions that the parameters c , b , and $\tilde{b}$ must satisfy. Are these conditions also sufficient for the containment relation between these two halfspaces?

Solution

Problem 8

For a pair of given sets $C, D \subseteq \mathbb{R}^n$, consider the following set

$$S = \{(a, b) \in \mathbb{R}^{n+1} \mid a^T x \leq b \text{ for all } x \in C, a^T x \geq b \text{ for all } x \in D\}.$$

Prove that the set S is convex.

Solution

Problem 9

Let $C = \{X \in \mathbb{R}^{n \times n} \mid a^T X a \geq 0, \forall a \in \mathbb{R}^n\}$. Determine whether the set

$$\mathcal{S} = \{A \in \mathbb{R}^{n \times n} \mid A - M \in C\}$$

is convex, where $M \in C$ is a fixed matrix. Provide justification or a counterexample. Note that usually the definition of a matrix being positive-definite requires the matrix to be symmetric, but this problem removes that condition.

Solution

Problem 10

Plot the norm balls of radius $r = 1$ for the following norms in $\mathbb{R}^2$ and $\mathbb{R}^3$:

- (a) ℓ_1 norm;
- (b) ℓ_∞ norm;
- (c) ℓ_2 norm.

Solution